

Ann Lia Sam

Integrated M.Sc. (Physics), National Institute of Science Education and Research (NISER), Bhubaneswar,
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Research Interests

Quantum computation and quantum information; quantum optics and cavity QED; light-matter interaction and molecular polaritonics; open quantum systems; quantum many-body and condensed matter physics.

Education

Integrated M.Sc. in Physics	2024 - Present
National Institute of Science Education and Research (NISER), Bhubaneswar	
Senior Secondary (Class XII), CBSE - Physics, Chemistry, Mathematics, Biology	2021 - 2023
St. Antony's Public School, Kottayam, Kerala	92%
Secondary (Class X), CBSE	2021
Holy Trinity Vidyabhavan, Karthikappally, Kerala	96%

Research Experience

Summer Research Intern - Indian Institute of Technology (IIT) Madras	June - July 2026
<i>Supervisors:</i> Prof. Krishna R. Nandipati and Prof. Athreya Shankar	
<i>Disorder in Cavity-Coupled Jahn-Teller Active Molecules</i>	
- Studied Jahn-Teller ($E \times e$) active molecules strongly coupled to the two circularly polarised modes of a Fabry-Pérot cavity, and reproduced the single-molecule cavity-JT polariton spectrum and the two-molecule <i>collective vibronic cascade</i> of the literature. - Extended the cavity-JT model to include <i>static energy disorder</i>, drawing each molecular transition energy from a Gaussian ensemble in the spirit of the disordered Holstein-Tavis-Cummings model, and interpreted the results through the random arrowhead-matrix picture of bright/dark-state mixing.	
Summer Research Intern - Indian Institute of Science (IISc), Bangalore	June - July 2025
<i>Supervisor:</i> Prof. Anoop Thomas	
<i>Vibrational Strong Coupling and its Effect on Photoinduced Electron Transfer</i>	
- Investigated vibrational strong coupling (VSC) and its influence on photoinduced electron transfer rates in molecular systems. - Fabricated Fabry-Pérot microcavities by spin-coating and characterised them through transmission spectra. - Analysed biexponential fluorescence decay kinetics using time-resolved spectroscopy. - Studied the hybridization of molecular vibrations with the cavity vacuum field and the resulting vibro-polaritonic states.	

Technical Skills

Programming	Python (NumPy, SciPy, Matplotlib, SymPy, QuTiP), $\LaTeX$, Git, Linux
Experimental	Microcavity fabrication and spin-coating; FTIR, UV-Vis, fluorescence and time-resolved spectroscopy; optical alignment and interferometry

Relevant Coursework & Laboratory Work

Completed	Quantum Mechanics I, Classical Mechanics I, Electromagnetism I & II, Mathematical Methods of Physics, Linear Optics (with laboratory), Electronics (with laboratory)
In Progress	Quantum Mechanics II, Statistical Mechanics, Computational Physics, General Physics Laboratory, Modern Physics Laboratory
Linear Optics Laboratory: Gaussian beam propagation and beam-waist measurement, interference and diffraction patterns, and interferometric characterisation of optical cavities.	

- **Modern Physics Laboratory:** foundational quantum and atomic physics experiments, with emphasis on measurement statistics and error analysis.
- **Computational Physics:** Gauss–Jordan elimination and LU decomposition for linear systems, Jacobi/Gauss–Seidel iterative solvers, root finding (Regula falsi, Newton–Raphson, Laguerre), numerical integration (Simpson, Monte Carlo), ODE solvers (Runge–Kutta, predictor–corrector) for initial- and boundary-value problems, and least-squares fitting.

Achievements & Scholarships

- All India Rank **397** in the National Entrance Screening Test (**NEST**) 2024.
- All India Rank **3148** in the IISER Aptitude Test (**IAT**) 2024.
- Recipient of the **DAE-DISHA** scholarship (Department of Atomic Energy, Government of India).

Additional Information

Languages	English and Malayalam (fluent); Hindi and Tamil (intermediate)
Interests	Reading, classical music(Carnatic), dance, playing the violin